



Mark Scheme

Mock Set 4

Pearson Edexcel GCE

In AS Level Mathematics (8MA0)

Paper 01 Pure Mathematics

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS
General Instructions for Marking

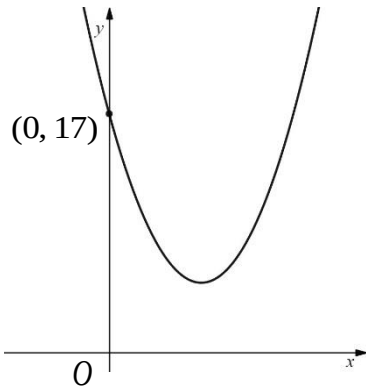
1. The total number of marks for the paper is 100.
2. The Edexcel Mathematics mark schemes use the following types of marks:
 - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
 - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
 - **B** marks are unconditional accuracy marks (independent of M marks)
 - Marks should not be subdivided.
3. Abbreviations
These are some of the traditional marking abbreviations that will appear in the mark schemes.
 - bod – benefit of doubt
 - ft – follow through
 - the symbol $\surd$ will be used for correct ft
 - cao – correct answer only
 - cso - correct solution only. There must be no errors in this part of the question to obtain this mark
 - isw – ignore subsequent working
 - awrt – answers which round to
 - SC: special case
 - oe – or equivalent (and appropriate)
 - dep – dependent
 - indep – independent
 - dp decimal places
 - sf significant figures
 - * The answer is printed on the paper
 - $\square$ The second mark is dependent on gaining the first mark
4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.
5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.
6. Ignore wrong working or incorrect statements following a correct answer.
7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternatives answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Question	Scheme	Marks	AOs
1	$4u^2 - u - 18 = 0 \Rightarrow (4u - 9)(u + 2) = 0 \Rightarrow u = \dots$	M1	1.1b
	$(u =) \frac{9}{4}, (-2)$	A1	1.1b
	$(x =) \pm \sqrt{\frac{9}{4}}$	M1	1.1a
	$(x =) \pm \frac{3}{2}$	A1	1.1b
		(4)	
(4 marks)			
Notes			
M1: Attempts to solve the quadratic $4u^2 - u - 18 = 0$ by either factorising, completing the square or using the quadratic formula. May be in terms of x e.g. $(4x^2 - 9)(x^2 + 2) = 0 \Rightarrow x^2 = \dots$ A1: $(u =) \frac{9}{4}, (-2)$ (do not be concerned with labelling) Do not withhold if the negative root is not seen. M1: Attempts to find the square root of their positive root. A1: $\pm \frac{3}{2}$ only Note: Answers only, with no working, would score no marks for this question.			

Question	Scheme	Marks	AOs
2(a)	$x^3 \rightarrow x^2$ or $x^2 \rightarrow x$ or $-10x \rightarrow -10$	M1	1.1b
	$\frac{dy}{dx} = 15x^2 + 19x - 10$	A1	1.1b
		(2)	
(b)	$\left(\frac{dy}{dx} =\right) 15(-2)^2 + 19(-2) - 10 = \dots (=12)$	M1	1.1b
	$y - 6 = "12"(x + 2)$	dM1	1.1b
	$y = 12x + 30$	A1	1.1b
		(3)	
(5 marks)			
Notes			
(a) M1: Reduces the power of x by one on one of the terms (indices do not need to be processed) A1: $15x^2 + 19x - 10$. Allow $19x^1$ (b) M1: Attempts to find the gradient of the curve at $x = -2$ dM1: Attempts to find the equation of the tangent at $(-2, 6)$ using their gradient. If they use $y = mx + c$ they must proceed as far as $c = \dots$. It is dependent on the first method mark. A1: $y = 12x + 30$			

Question	Scheme	Marks	AOs
3	$\frac{4-\sqrt{6}}{5\sqrt{2}+4\sqrt{3}} \times \frac{5\sqrt{2}-4\sqrt{3}}{5\sqrt{2}-4\sqrt{3}} = \frac{20\sqrt{2}-16\sqrt{3}-5\sqrt{12}+4\sqrt{18}}{50-48}$	M1	2.1
	$\frac{32\sqrt{2}-26\sqrt{3}}{2}$	dM1	1.1b
	$16\sqrt{2}-13\sqrt{3}$	A1	1.1b
		(3)	
Alt 3	$4-\sqrt{6} = \dots a + \dots b + (\dots a + \dots b)\sqrt{6}$	M1	2.1
	$4=10a+12b \text{ and } -1=4a+5b \Rightarrow a=\dots \text{ or } b=\dots$	dM1	1.1b
	$16\sqrt{2}-13\sqrt{3}$	A1	1.1b
(3 marks)			
Notes			
M1: Multiplies the numerator and denominator by $5\sqrt{2}-4\sqrt{3}$ (or a multiple of this) and proceeds to $\frac{\dots}{50-48}$ or $\frac{\dots}{2}$ or $\frac{20\sqrt{2}-16\sqrt{3}-5\sqrt{12}+4\sqrt{18}}{\dots}$ condoning slips when multiplying out including signs. dM1: Attempts to simplify their fraction so that all the terms on the numerator are in terms of $\sqrt{2}$ or $\sqrt{3}$. E.g. $\frac{32\sqrt{2}-26\sqrt{3}}{2}$. Do not allow this mark for proceeding straight to the final answer. Note that if they proceed straight to $\frac{32\sqrt{2}-26\sqrt{3}}{2}$ with no other working seen then this scores M1dM0A0 as they have not attempted to simplify their fraction (i.e., we have not seen terms that they have tried to collect) A1: $16\sqrt{2}-13\sqrt{3}$			
Alternative method			
M1: Multiplies both sides by $5\sqrt{2}+4\sqrt{3}$ and proceeds to $4-\sqrt{6} = \dots a + \dots b + (\dots a + \dots b)\sqrt{6}$ which may be unsimplified. (The correct equation is $4-\sqrt{6} = 10a+12b+(4a+5b)\sqrt{6}$) dM1: Attempts to form two simultaneous equations by equating coefficients using their equation and proceeds to find a value for a or b (condoning slips). e.g. $4=10a+12b$ and $-1=4a+5b \Rightarrow a=\dots$ or $b=\dots$ which may be unsimplified A1: $16\sqrt{2}-13\sqrt{3}$			

Question	Scheme	Marks	AOs
4(a)(i) (ii)	$x = 2$	B1	1.2
	$\left(0, -\frac{1}{2}\right)$	B1	1.1b
		(2)	
(b)(i) (ii)	$(3, 5)$	B1	1.1b
	$\left(\frac{3}{2}, 1\right)$	B1	1.1b
		(2)	
(4 marks)			
Notes			
(a) (i) B1: $x = 2$ (ii) B1: $\left(0, -\frac{1}{2}\right)$ o.e. (b) (i) B1: $(3, 5)$ o.e. (ii) B1: $\left(\frac{3}{2}, 1\right)$ o.e.			

Question	Scheme	Marks	AOs
5(a)	$a = 3$	B1	1.1b
	$(x \pm 2)^2$	M1	1.1b
	$3(x - 2)^2 + 5$	A1	1.1b
		(3)	
(b)	(2, 5)	B1ft	2.2a
		(1)	
(c)		B1B1	1.1b 1.1b
		(2)	
(6 marks)			
Notes			
(a) B1: $a = 3$ or $3(x \pm \dots)^2$ M1: $(x \pm 2)^2$ A1: $3(x - 2)^2 + 5$ (b) Mark (b) and (c) together B1ft: (2, 5) follow through on their part (a) so score for $(-b, c)$ May be seen on their sketch. (c) B1: Correct positive quadratic shape with a minimum in the first quadrant B1: y-intercept at (0, 17). Condone 17 marked on the y-axis for this mark			

Question	Scheme	Marks	AOs
6	$3^{3(2x-3)} = \frac{1}{3^2 \times 3^{\frac{1}{2}}}$	M1	3.1a
	$3^{6x-9} = 3^{-\frac{5}{2}} \Rightarrow 6x-9 = -\frac{5}{2} \Rightarrow x = \dots$	M1	1.1b
	$x = \frac{13}{12}$	A1	1.1b
		(3)	
(3 marks)			
Notes			
M1: Attempts to write each number in base 3. May alternatively write each number in another base e.g. 9 M1: Manipulates the equation to $3^{\dots} = 3^{\dots}$ or $3^{\dots} = 1$, proceeds to a linear equation in x and attempts to solve. (Alternatively may manipulate the equation to e.g. $9^{\dots} = 9^{\dots}$ first) Condone slips on expanding the brackets before proceeding to a linear equation in x. A1: $x = \frac{13}{12}$ or exact equivalent			

Question	Scheme	Marks	AOs
7(a)	Uses or implies that $P = an + b$	B1	3.3
	$a = \frac{65 - 24}{20} = 2.05$	M1	3.1b
	e.g. $24 = 10 \times "2.05" + b \Rightarrow b = \dots$	M1	1.1b
	$P = 2.05n + 3.5$	A1	2.2a
		(4)	
(b)	$P = 2.05 \times 40 + 3.5 = 85.5$	M1	3.4
	85 500	A1	3.2a
		(2)	
(c)	$85500 \approx 85000$ so this is a good model	B1ft	3.5a
		(1)	
(7 marks)			
Notes			
(a) B1: Uses or implies that $P = an + b$ (may use other letters for a and b) M1: Attempts to find a by finding the gradient between (10, 24) and (30, 65). Score for the expression $\frac{65 - 24}{20}$. Alternatively, forms two simultaneous equations $24 = 10a + b$ and $65 = 30a + b$ and solves to find a or b condoning arithmetical slips. Condone the use of 24 000 and 65 000 which can score maximum B1M1M1A0 M1: Uses their value for a, substitutes into $P = an + b$ with either (10, 24) or (30, 65) and proceeds to find a value for b. The coordinates must be substituted into the equation the correct way round. Alternatively, using simultaneous equations, proceeds to find a value for a and for b, condoning arithmetical slips. A1: $P = 2.05n + 3.5$ (b) M1: Substitutes $n = 40$ into their equation found in (a) to find a value for P A1: 85 500 o.e. (which may have come from $P = 2050n + 3500$) (c) B1ft: States that it is a good model because 85 500 is approximately equal to 85 000. Follow through on their answer to (b) provided the value is close to 85 000 (allow values between 80 000 and 90 000). Allow comparison of 85.5 and 85. Must conclude this is a good model.			

Question	Scheme	Marks	AOs
8(a)	$f(2) = 5(2)^3 - 7(2)^2 - 8(2) + a = 0$ or $ \begin{array}{r} 5x^2 + 3x - 2 \\ x-2 \overline{) 5x^3 - 7x^2 - 8x + a} \\ \underline{5x^3 - 10x^2} \\ 3x^2 - 8x \\ \underline{3x^2 - 6x} \\ -2x + a \\ \underline{-2x + 4} \Rightarrow a - 4 = 0 \end{array} $	M1	2.1
	$5 \times 8 - 7 \times 4 - 8 \times 2 + a = 0 \Rightarrow a = 4^*$ or $a - 4 = 0 \Rightarrow a = 4^*$	A1*	1.1b
		(2)	
(b)	$5x^3 - 7x^2 - 8x + 4 = (x-2)(5x^2 + 3x - 2)$	M1 A1	1.1b 1.1b
	$(x-2)(5x-2)(x+1) = 0 \Rightarrow x = 2, \frac{2}{5}, -1$	dM1 A1	1.1b 1.1b
		(4)	
(6 marks)			
Notes			
(a) M1: Attempts $f(2) = 0$, $x = 2$ embedded in the equation set equal to 0 is sufficient. The $=0$ may be implied by further work. May also attempt to divide $f(x)$ by $x-2$, proceeding as far as a quadratic quotient and a remainder in the form $a \pm \dots$ which is set equal to 0. A1*: Achieves $a = 4$ with at least one intermediate stage of working and no errors seen, including invisible brackets. If via algebraic division, the quadratic quotient must be $5x^2 + 3x - 2$ (b) M1: Attempts to find the quadratic factor either via division or inspection. If via division score for a quadratic factor $(5x^2 \pm 3x \pm \dots)$ or via inspection $(5x^2 \pm \dots x \pm 2)$ A1: $(5x^2 + 3x - 2)$ which may be seen as the quotient if carrying out algebraic division. Does not need to be written with $(x-2)$ dM1: Attempts to factorise their quadratic, completes the square or uses the formula to solve their quadratic. Usual rules apply. It is dependent on a quadratic factor having been found. They cannot proceed from the cubic directly to a fully factorised expression. A1: $2, \frac{2}{5}, -1$ only Note: Answers only, with no working, would score no marks for part (b).			

Question	Scheme	Marks	AOs
9	$3 = \log_3 27$ or $2 \log_3 \sqrt{x} = \log_3 x$	M1	1.1b
	e.g. $\log_3(2x+3)(x-4) = \log_3 27x$	M1	1.1b
	$(2x+3)(x-4) = 27x$	A1	2.1
	$2x^2 - 32x - 12 = 0 \Rightarrow x^2 - 16x - 6 = 0 \Rightarrow x = \dots$	M1	1.1b
	$8 + \sqrt{70}$ only	A1	2.3
		(5)	
(5 marks)			
Notes			
M1: Uses $3 = \log_3 27$ or $2 \log_3 \sqrt{x} = \log_3 x$ (may be implied) M1: Uses the addition or subtraction law of logarithms at least once A1: A correct equation with logarithms correctly removed e.g. $(2x+3)(x-4) = 27x$. Do not allow this mark for e.g. $\frac{\log_3(2x+3)(x-4)}{\log_3 27x} = 0 \Rightarrow \frac{(2x+3)(x-4)}{27x} = 1$ where the correct equation came from incorrect working. M1: Attempts to solve their three-term quadratic (usual rules apply for solving a quadratic). If the root is via a calculator you will need to check this. Accept rounded decimals to 3sf. A1: $8 + \sqrt{70}$ only			

Question	Scheme	Marks	AOs
10(a)	$17 + 25e^{0.08 \times T} = 62 \Rightarrow e^{0.08 \times T} = \dots$	M1	3.4
	$e^{0.08 \times T} = \frac{48}{25} \Rightarrow T = \frac{\ln\left(\frac{48}{25}\right)}{0.08}$	M1	1.1b
	$T = 8.2$ (months)	A1	2.2a
		(3)	
(b)	$\frac{dN}{dt} = 0.08 \times 25e^{0.08t}$	M1	3.3
	$\frac{dN}{dt} = 0.08 \times 25e^{0.08 \times 24} = 13.6$	dM1	1.1b
	13.6 million users per month	A1	3.4
		(3)	
(c)	The model increases exponentially so not suitable for large values of t	B1	3.5b
		(1)	
(7 marks)			
Notes			
(a) M1: Sets $17 + 25e^{0.08 \times T} = 62$ and rearranges to $e^{0.08 \times T} = \dots$ M1: Proceeds from an expression of the form $Ae^{0.08 \times T} = B$ where $A, B > 0$, takes lns of both sides and proceeds to find an expression for T A1: awrt 8.2 (months) (b) M1: Attempts to differentiate with respect to t. Score for an expression of the form $\dots e^{0.08t}$. Do not allow for $\frac{dN}{dt} = 17 + 25e^{0.08t}$ dM1: Substitutes in $t = 24$ and proceeds to find a value for $\frac{dN}{dt}$ A1: 13.6 million users per month o.e. Must have units. Do not condone 13.6 for this mark. (c) B1: E.g. the model suggests that the number of users would increase without bound which is unrealistic / the number of users is unlikely to keep increasing exponentially.			

Question	Scheme	Marks	AOs
11(a)	$A = 3^7 = 2187$	B1	1.1b
		(1)	
(b)	${}^7C_1 \times 3^6 \times \frac{k}{9}$ or ${}^7C_2 \times 3^5 \times \left(\frac{k}{9}\right)^2 = 406$	M1	1.1b
	${}^7C_1 \times 3^6 \times \frac{k}{9} + {}^7C_2 \times 3^5 \times \left(\frac{k}{9}\right)^2 = 406 \Rightarrow \dots k^2 \pm \dots k \pm \dots = 406$	dM1	1.1b
	$\Rightarrow 9k^2 + 81k - 58 = 0 *$	A1*	1.1b
		(3)	
(c)(i)	$k = \frac{2}{3}$	B1	1.1b
(ii)	$"63" \times \left(\frac{2}{3}\right)^2 = 28$	M1 A1	1.1b 1.1b
		(3)	
(d)	$2 \times "2187" + 2 \times "28" x^2$	M1	3.1a
	$= 4374 + 56x^2$	A1	1.1b
		(2)	
(9 marks)			
Notes			
(a) B1: 3^7 or 2187 (b) M1: Attempts to find either the coefficient of x or the coefficient of x^2. Condone invisible brackets. May be seen in the equation $B + C = 406$ dM1: Substitutes their coefficients for x and x^2 into the equation $B + C = 406$ and proceeds to a three-term quadratic in k which is not the given answer. Terms do not need to be on the same side. It is dependent on the previous method mark. A1*: Achieves the given answer with at least one intermediate stage of working seen between their equation for $B + C = 406$ and proceeds to the answer. e.g. ${}^7C_1 \times 3^6 \times \frac{k}{9} + {}^7C_2 \times 3^5 \times \left(\frac{k}{9}\right)^2 = 406 \Rightarrow 567k + 63k^2 = 406 \Rightarrow 9k^2 + 81k - 58 = 0$ (c) (i) B1: $k = \frac{2}{3}$ only (must reject the negative root if found) (ii) M1: Substitutes their positive value for k into their expression for C in terms of k from part (ii) and proceeds to find a value A1: 28 (d) M1: Attempts $2 \times A + 2 \times C$ using their A and their C A1: $4374 + 56x^2$ cao (provided both previous marks have been scored)			

Question	Scheme	Marks	AOs
12(a)	$3\sin\theta \frac{\sin\theta}{\cos\theta} = 4$	M1	1.1b
	$3(1 - \cos^2\theta) = 4\cos\theta$	M1	1.1b
	$3\cos^2\theta + 4\cos\theta - 3 = 0^*$	A1*	2.1
		(3)	
(b)	$3\cos^2 2x + 4\cos 2x - 3 = 0 \Rightarrow \cos 2x = \frac{-4 \pm \sqrt{16 - 4 \times 3 \times (-3)}}{2 \times 3}$	M1	1.1b
	$(\cos 2x =) \frac{-2 + \sqrt{13}}{3}$	A1	2.3
	$x = \frac{\cos^{-1}\left(\frac{-2 + \sqrt{13}}{3}\right)}{2}$	dM1	1.1b
	awrt 28.8° , awrt 151.2°	A1	1.1b
		(4)	
(7 marks)			
Notes			
(a) M1: Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ M1: Uses $\pm\sin^2\theta \pm \cos^2\theta = \pm 1$ to achieve an equation in terms of $\cos\theta$ A1*: Achieves the given answer $3\cos^2\theta + 4\cos\theta - 3 = 0$ with no errors seen including invisible brackets (b) M1: Attempts to solve $3\cos^2 2x + 4\cos 2x - 3 = 0$ by either completing the square or using the quadratic formula or using a calculator. A1: Deduces $(\cos 2x =) \frac{-2 + \sqrt{13}}{3}$ (the other root if found must be rejected / no other solutions found) dM1: Attempts to find an angle in the given range for their solution to the quadratic using the correct processing. May be implied by awrt 29 or awrt 151. A1: awrt 28.8°, awrt 151.2° and no others in the range. The previous three marks must have been scored (answers with no working score no marks).			

Question	Scheme	Marks	AOs
13(a)	$7\mathbf{i} + 5\mathbf{j} + 2(7-1)\mathbf{i} + 2(5+3)\mathbf{j}$	M1	3.1a
	$= 19\mathbf{i} + 21\mathbf{j}$	A1	1.1b
		(2)	
(b)	$ \overrightarrow{AD} = \sqrt{17} = \sqrt{1^2 + (p-1)^2} \Rightarrow p = \dots$	M1	2.1
	$p = -3, (5)$	A1	1.1b
	$ \overrightarrow{AB} = \sqrt{6^2 + 8^2} = 10$ and $ \overrightarrow{DB} = \sqrt{(7-2)^2 + (5-(-3-4))^2} = 13$	M1	2.2a
	$\cos DAB = \frac{17 + 100 - 169}{2 \times \sqrt{17} \times 10} \Rightarrow \text{angle } DAB = \dots$	dM1	3.1a
	angle $DAB = \text{awrt } 129.1^\circ$	A1	1.1b
		(5)	
(7 marks)			
Notes			
(a) M1: Complete attempt to find the position vector of C A1: $19\mathbf{i} + 21\mathbf{j}$ accept column vector notation (b) M1: Attempts to form the equation using the magnitude of AD and proceeds to find a value for p. May be implied by sight of $(2, -7)$ A1: -3 (ignore the presence or absence of 5). Implied by $(2, -7)$ which may appear on a diagram. M1: Deduces $\overrightarrow{BD}$ needs to be as big as possible so uses their $p = -3$ and attempts to find the magnitude of both AB and DB. Their value for p must be the one which gives the greater magnitude for BD. You may not see any reference to the other value for p if it was rejected when found. May be implied by further work. dM1: Attempts the cosine rule with the lengths in the correct places and proceeds to find angle DAB. Condone slips. It is dependent on the previous method mark. May work in radians which is acceptable. A1: awrt 129.1 (must be in degrees)			

Question	Scheme	Marks	AOs
14(a)(i) (ii)	$(2k, -k)$	B1	1.1b
	$(\pm 2k)^2 + (\pm k)^2 = \dots$	M1	1.1b
	$\sqrt{5k^2 - 80}$	A1	1.1b
		(3)	
(b)	$x^2 + (k - 2x)^2 - 4kx + 2k(k - 2x) + 80 = 0$ $\Rightarrow 5x^2 - 12kx + 3k^2 + 80 = 0$	M1A1	3.1a 1.1b
	$(-12k)^2 - 4 \times 5 \times (3k^2 + 80) < 0 \Rightarrow CV = \frac{20\sqrt{21}}{21}$	M1	2.1
	$\left\{ k : 4 < k < \frac{20\sqrt{21}}{21} \right\}$	M1A1	1.1b 2.5
		(5)	
(8 marks)			
Notes			
(a) B1: $(2k, -k)$ M1: Attempts to use Pythagoras' theorem to find the radius² (may be implied by their radius) A1: $\sqrt{5k^2 - 80}$ (b) M1: Substitutes $y = k - 2x$ into the equation for C, multiplies out and proceeds to a three term quadratic in x A1: $5x^2 - 12kx + 3k^2 + 80 = 0$ or simplified equivalent. Terms do not need to be on the same side of the equation and the = 0 may be implied. May also be implied by their values for a, b and c when using the discriminant. M1: Attempts the discriminant using their a, b and c where b and c are both in terms of k and attempts to find the critical value which is greater than 4 (usual rules apply for solving a quadratic). Do not be concerned if they find the other critical value "$-\frac{20\sqrt{21}}{21}$" for this mark. M1: Attempts to find the correct inside region using their critical value which is greater than 4 so score for either $4 < k < \frac{20\sqrt{21}}{21}$ or "$-\frac{20\sqrt{21}}{21} < k < \frac{20\sqrt{21}}{21}$". Set notation is not required for this mark. A1: $\left\{ k : 4 < k < \frac{20\sqrt{21}}{21} \right\}$ or equivalent using set notation. Look for $\{ \}$ with the correct region in terms of k.			

Question	Scheme	Marks	AOs
15(a)	$-\frac{2}{5}$	B1	1.1b
		(1)	
(b)	$-\frac{2}{5} \rightarrow \frac{5}{2}$	M1	1.1b
	$y - 16 = \frac{5}{2}(x - 10)$	M1	1.1b
	$5x - 2y - 18 = 0$	A1	2.2a
		(3)	
(c)	$5x - 18 = 0 \Rightarrow x = \frac{18}{5}$	M1 A1	1.1b 1.1b
		(2)	
(d)	Coordinates of R (50, 0)	B1	2.2a
	$\text{Area} = \frac{1}{2} \times 16 \times \left(50 - \frac{18}{5}\right) = \frac{1856}{5} (=371.2) \text{ (units}^2\text{)}$	M1 A1	3.1a 1.1b
		(3)	
(e)	Curve crosses positive x axis at $x = 20$ only	B1	2.2a
	$\int \left(-\frac{3}{50}x^2 + \frac{1}{5}x + 20\right) dx = -\frac{1}{50}x^3 + \frac{1}{10}x^2 + 20x (+c)$	M1A1	1.1b 1.1b
	$\left(-\frac{1}{50}(20)^3 + \frac{1}{10}20^2 + 20 \times 20\right) - \left(-\frac{1}{50} \times 10^3 + \frac{1}{10}10^2 + 20 \times 10\right)$ (=90)	M1	1.1b
	$\frac{\frac{1}{2} \times 16 \times (50 - 10) - "90"}{371.2} = \frac{575}{928}$	dM1 A1	3.1a 2.2a
		(6)	

(15 marks)

Notes

(a)

B1: $-\frac{2}{5}$ o.e.

(b)

M1: Finds the negative reciprocal of their gradient in (a)

M1: Attempts the equation of the line using their changed gradient from (a) and (10, 16), with both coordinates substituted in correctly e.g. $y - 16 = \frac{5}{2}(x - 10)$. If they use $y = mx + c$ they must proceed as far as $c = \dots$

A1: $5x - 2y - 18 = 0$ or any integer multiple

(c)

M1: Sets $y = 0$ for their linear equation found in (b) and proceeds to find a value for x

A1: $x = \frac{18}{5}$ o.e.

(d)

B1: Deduces that the x coordinate of R is 50

M1: Attempts to find the area of triangle PQR using their x coordinate for R and their x coordinate for Q

A1: 371.2 o.e. (units²)

(e)

B1: Deduces that the curve crosses the positive x axis at 20 (may be via a calculator)

M1: Attempts to integrate the equation of the curve. Look for one index correct

A1: $-\frac{1}{50}x^3 + \frac{1}{10}x^2 + 20x$ with or without the constant of integration

M1: Attempts to substitute in the limits “20” and 10 and subtracts either way round

dM1: Attempts to find the fraction of triangle PQR shaded. It is dependent on the previous two method marks.

A1: $\frac{575}{928}$ cao

Question	Scheme	Marks	AOs
16	Attempts to find $n^3 + 3n^2 - 2n$ for either odds or evens e.g. $n = 2k$ $(2k)^3 + 3(2k)^2 - 2(2k) = \dots$ or e.g. $n = 2k + 1$ $(2k + 1)^3 + 3(2k + 1)^2 - 2(2k + 1) = \dots$	M1	2.1
	e.g. $8k^3 + 12k^2 - 4k = 2(4k^3 + 6k^2 - 2k)$ which is even or e.g. $8k^3 + 24k^2 + 14k + 2 = 2(4k^3 + 12k^2 + 7k + 1)$ which is even	A1	2.2a
	Attempts to find $n^3 + 3n^2 - 2n$ for odds and evens e.g. $n = 2k$ $(2k)^3 + 3(2k)^2 - 2(2k) = \dots$ and e.g. $n = 2k + 1$ $(2k + 1)^3 + 3(2k + 1)^2 - 2(2k + 1) = \dots$	dM1	3.1a
	e.g. $8k^3 + 12k^2 - 4k = 2(4k^3 + 6k^2 - 2k)$ and e.g. $8k^3 + 24k^2 + 14k + 2 = 2(4k^3 + 12k^2 + 7k + 1)$ hence even for all positive integers n *	A1*	2.4
		(4)	
(4 marks)			
Notes			
There should be no errors in the algebra but allow e.g. invisible brackets to be “recovered”. Condone the use of e.g. $n = 2n$ or $n = 2n + 1$ for the first three marks. M1: Uses algebra to describe odds e.g. $n = 2k$ or evens e.g. $n = 2k + 1$ and attempts $n^3 + 3n^2 - 2n$ (allow equivalent representation of odd or even e.g. $n = 2k - 1$ or $n = 2k + 2$). They must attempt to expand the brackets to achieve a cubic expression but condone arithmetical slips. Note they may factorise the cubic first $n^3 + 3n^2 - 2n = n(n^2 + 3n - 2)$ before attempting to describe odds or evens. With this approach it is sufficient for the even case to just say that as n is even then the expression will be even. A1: They must achieve a correct expanded cubic expression (which may be factorised) for either odds or evens and shows or gives an explanation as to why the expression is even. e.g. $8k^3 + 12k^2 - 4k = 2(4k^3 + 6k^2 - 2k)$ which is even e.g. $8k^3 + 12k^2 - 4k$ which is even as 2 is a factor of each term but allow the even case of e.g. $n^3 + 3n^2 - 2n = n(n^2 + 3n - 2)$ which is even if n is even dM1: Uses algebra to describe odds e.g. $n = 2k$ and evens e.g. $n = 2k + 1$ and attempts $n^3 + 3n^2 - 2n$ (allow equivalent representation of odd or even e.g. $n = 2k - 1$ and $n = 2k + 2$). They must attempt to expand the brackets to achieve a cubic expression but condone arithmetical slips. Note they may factorise the cubic first $n^3 + 3n^2 - 2n = n(n^2 + 3n - 2)$ before attempting to describe odds or evens. With this approach it is sufficient for the even case to just say that as n is even then the expression will be even.			

A1*: Fully correct proof. They must achieve a correct expanded $n^3 + 3n^2 - 2n$ for both odds and evens and shows or gives reasons/explanations as to why the expressions are even.
There must be an overall conclusion as well e.g. hence even for all positive integers
Withhold this mark for use of e.g. $n = 2n$ **or** $n = 2n + 1$